A project Work on **Internet of Things**

Entitled

Home Automation System

This project report is submitted in the partial fulfillment of the requirement provided for the award of Information Technology 'O' Level under NIELIT

Submitted by : **Suchismita Roy**

Registration No: 1472349

Under the Esteemed guidance of :

Mr. Shamik Gupta Bhaya

National Institute of Electronics and Information technology

(An Autonomous Scientific Society of Dept. of Electronics and Information Technology)
Ministry of Communications and Information Technology, Government of India

Silchar
Tarapur,Shibbari Road -788003
Year-2024

Certificate

This is to certify that Miss Suchismita Roy, (Registration number – 1472349) an IT ‘O’ level student of the National Institute of Electronics and Information Technology (NIELIT), Silchar Extension Center, worked under my guidance and support for her project on Internet of Things titled, “Home Automation System using Internet of Things” for her partial fulfillment of the award of Information Technology ‘O’ Level.

Place: Silchar,Shibbari Road

Date:

Mr. Shamik Gupta Bhaya

NIELIT Silchar EC

College Road-788004

Declaration

I, Suchismita Roy, bearing Registration No. 1472349, hereby declare that the subject matter of the project entitled on “Home Automation System Using Internet Of Things”, which is being submitted in fulfillment of IT ‘O’ Level under NIELIT. The Project Work has been carried out by me at the Institute itself. The result of the work carried out by me, that the contents of this Project did not form the basis for the award of any degree to me or to anybody else to the best of my knowledge. The dissertation has not been submitted in any other University/ Institute.

Place: NIELIT Silchar EC

Suchismita Roy

Date:

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Suchismita Roy

IT ‘O’ Level

NIELIT Silchar EC

Abstract

Due to the rapid development in the field of the Automation industry, human life is becoming more advanced and better in all aspects. In the present scenario, automated systems are being preferred over the non-automated system. With the rapid growth in the number of consumers using the internet over the past years, the internet has become an important part of life and IoT is the newest and emerging internet technology. Internet of things plays an important role in the human life because they are able to provide information and complete the given tasks while we are busy doing some other work.

In this project, I am going to implement a few such examples of the uses of Internet of Things in our day to day lives to make our life a little stress free.

With that, let's take into consideration some of the common issues faced by us in our homes along with their solutions:

1. Automated Security System.

2. Automatic fan operator by detecting temperature in the room.

3. Automatic switching on of bulb by detecting lack of brightness.

4. Automated alarm system by Smoke detector.

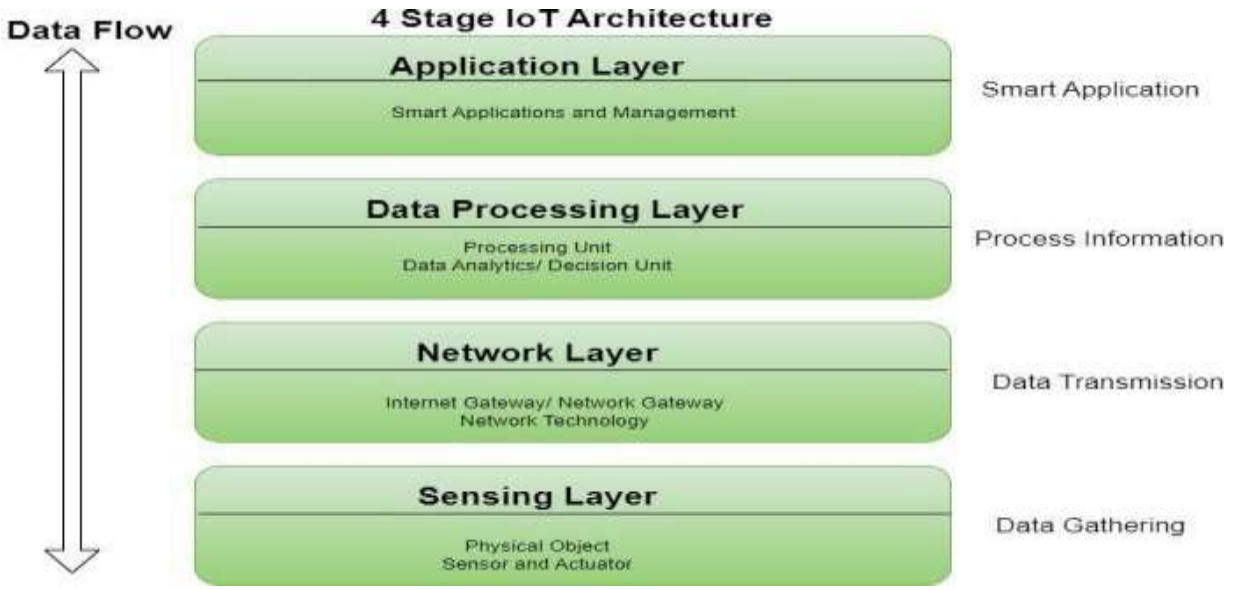
Introduction to IoT

IoT stands for Internet of Things. It refers to the interconnectedness of physical devices, such as appliances and vehicles , that are embedded with software, sensors, and connectivity which enables these objects to connect and exchange data. This technology allows for the collection and sharing of data from a vast network of devices, creating opportunities for more efficient and automated systems.

Internet of Things (IoT) is the networking of physical objects that contain electronics embedded within their architecture in order to communicate and sense interactions amongst each other or with respect to the external environment.

Architecture

The architecture of IoT is divided into 4 different layers i.e. Sensing Layer, Network Layer, Data processing Layer, and Application Layer as shown below.



Introduction to my Project

The name of my project is:

"Smart Home Automation Using Internet of Things ".

In this project we will see how we can make our lives much easier by escaping the little tasks which we need to do apart from the hectic workloads.

Aims and Objectives

1. An easier and stress-free lifestyle.
2. A tighter home security.
3. Alarm against fire accidents
4. Convenient and Smart operation.
5. Cost effective.
6. Easy to install.

Tools Used

Software:

Arduino IDE

Hardware:

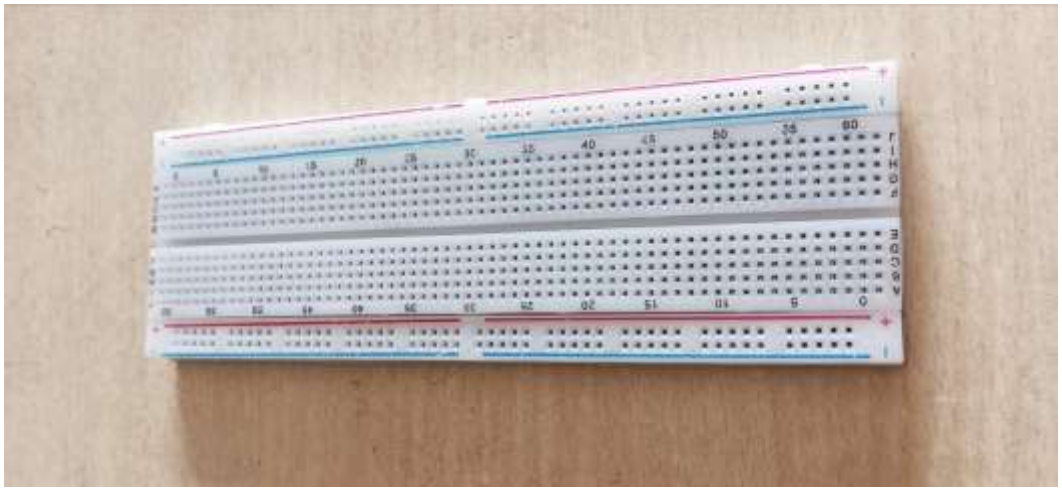
1. Arduino Uno R3
2. Breadboard
3. Jumper Wire Connectors.
4. DC Motor with Fan.
5. DHT11 Temperature and Humidity sensor.
6. HC-SR04 Ultrasonic Sensor.
7. DC 5V TMB12A05 Sound Buzzer
8. KY-018 LDR Light Sensor Module
9. LED bulb
10. Rectifier Diode, Resistor, Transistor
11. MQ2 Gas Sensor.

Components description

1. Arduino Uno R3

The Arduino Uno R3 is a microcontroller board based on a removable, dual-inline-package (DIP) ATmega328 AVR microcontroller. It has 20 digital input/output pins (of which 6 can be used as PWM outputs and 6 can be used as analog inputs). Programs can be loaded on to it from the easy-to-use Arduino computer program.

2. Breadboard

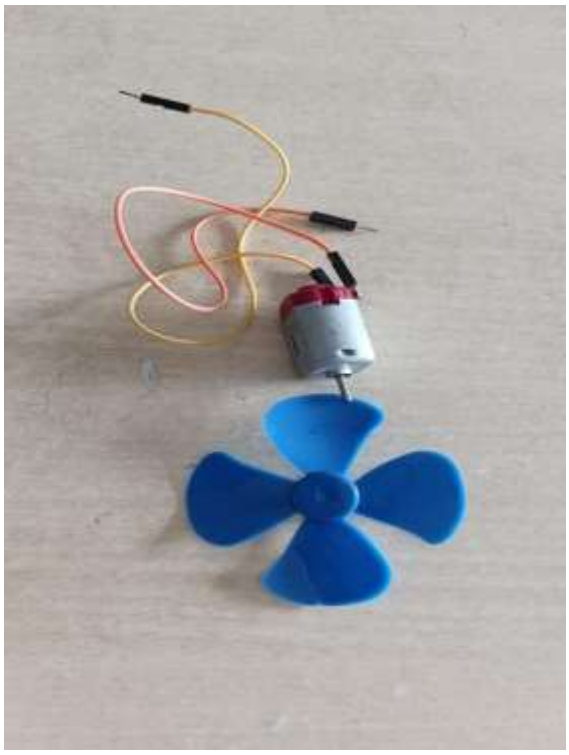


A breadboard (sometimes called protoboard) is essentially the foundation to construct and prototype electronics. A breadboard allows for easy and quick creation of temporary electronic circuits or to carry out experiments with circuit design

3. Jumper Wire Connectors

Jumper wires are used for making connections between items on our breadboard and our Arduino's header pins. We use them to wire up all our circuits.

4. DC Motor with Fan



A direct current, or DC, motor is the most common type of motor. DC motors normally have just two leads, one positive and one negative. If you connect these two leads directly to a battery, the motor along with the fan will rotate.

5. DHT11 Temperature and Humidity Sensor

The DHT11 is a basic, ultra low-cost digital temperature and humidity sensor. It uses a capacitive humidity sensor and a thermistor to measure the surrounding air and spits out a digital signal on the data pin (no analog input pins needed).

6. HC-SR04 Ultrasonic Sensor

HC-SR04 stands for High-Conductance ultrasonic sensor, which consists of a transmitter and receiver. The sensor measures how far things are without touching them, and it uses sound waves to get the measurements right.

7. DC 5V TMB12A05 Sound Buzzer

This voltage buzzer generally use in alarm circuit, communication equipment, piezo, Automobile electronics. This product normally operate on 5v dc and its current rating is less than equal to 30maA

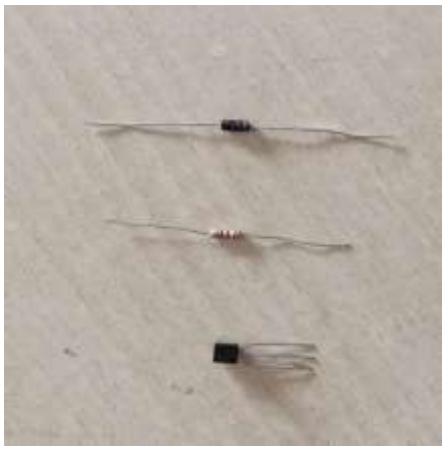
8. KY-018 LDR Light Sensor Module

The KY-018 light detection sensor module is a device that allows you to detect the presence or absence of light using an Arduino or other microcontroller. It consists of a light-dependent resistor (LDR) and a voltage divider circuit that outputs a voltage proportional to the amount of light detected.

9. LED Bulb

The LED, also known as a Light Emitting Diode, is a component that emits light.

10. Rectifier Diode, Resistor, Transistor



Rectifier diodes are mainly used for rectification which means to change AC to DC. They are used in circuits wherever a huge current should flow throughout the diode.

Resistor is used to resistance the flow of current. When resistor is placed in a circuit, the current flow decreases when current passes through the resistor.

Transistor is a semiconductor device used to amplify or switch electrical signals and power. It is one of the basic building blocks of modern electronics.

11. MQ2 Gas Sensor

The MQ-2 is a smoke and combustible gas sensor from Winsen. It can detect flammable gas in a range of 300 - 10000ppm. It's most common use is domestic gas leakage alarms and detectors with a high sensitivity to propane and smoke.

The Features of Home Automation

1. Automated Security System - This system is aimed towards a tighter security system in our house, for which I have made an Automated Alarm System which rings as soon as someone comes near any window or any door of the house which is an entrance.

2. Automated Fan Operator - When the temperature of the room is increased to a certain level,the fan can be automatically turned on without the need of switching it on manually. This system can help maintain the temperature of the place when there is no one to supervise the control of the Fan.

3. Automated Light Bulb operator - When the room reaches a certain Darkness level, the bulb is lit automatically without having to switch it on manually.

4. Smoke detection alarm - Whenever there is a fire or some suspicious smoke or gas in the room, the alarm is on and therefore we get informed about the presence of smoke in the room.

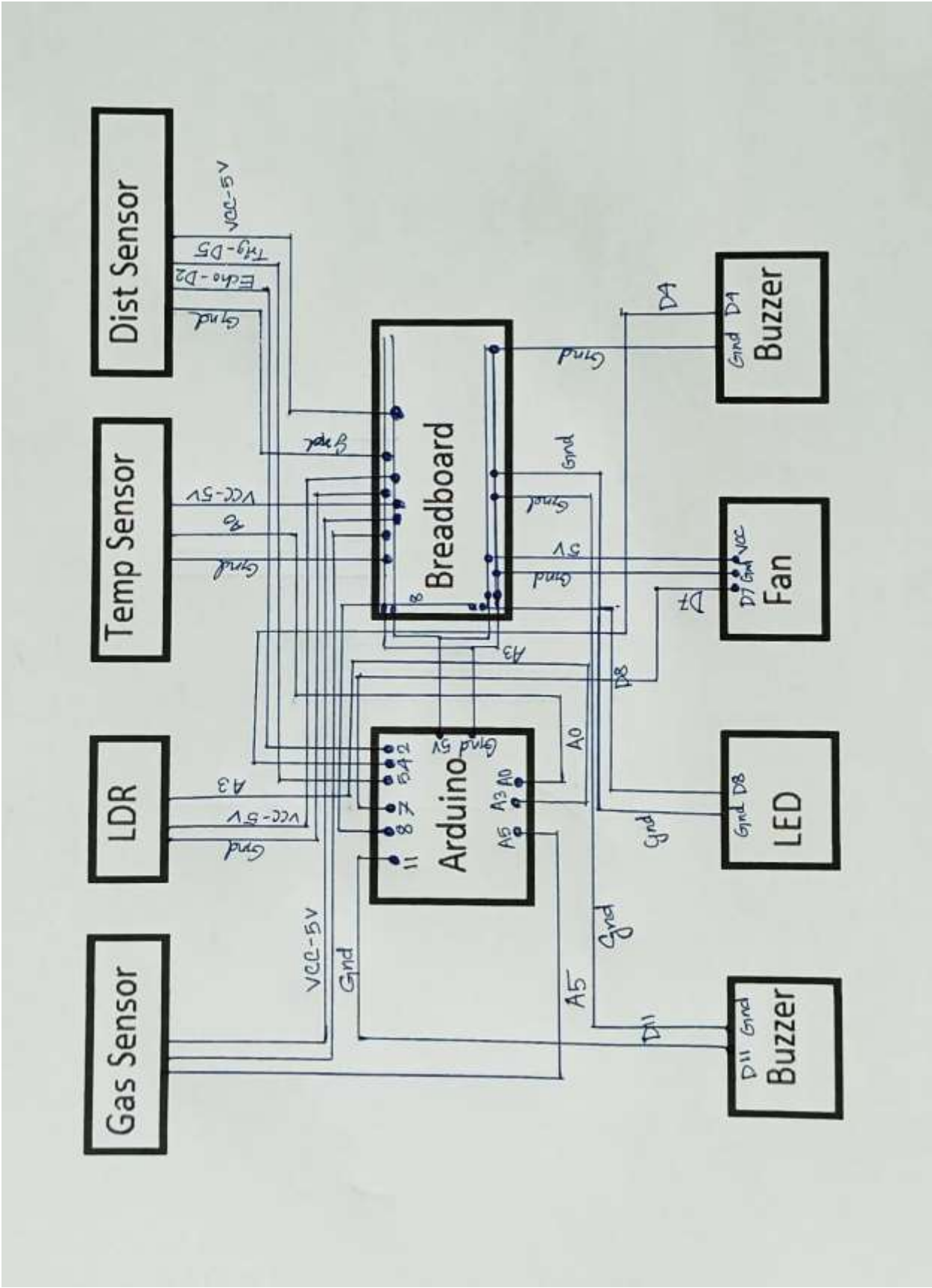
How does it Work?

In this project we entirely use the concept of Internet things i.e., in simple words connecting things with internet.

For, example we use different sensors and output devices and also monitor or control the changes in the outputs with respect to the inputs we obtained with the help of a code which we upload into the Arduino Uno R3

1. **Automated Security System:**
   In this system, we use an ultrasonic distance sensor named ***HC-SR04***, which detects any object at a certain distance. Here we set the distance limit upto ***20 cm***. If someone comes within the range of 20 cm ,the ***buzzer*** starts ringing which alarms us about someone entering the house.
2. **Automated Fan Operator:**
   In this system, we use a temperature and humidity sensor named ***DHT11*** Temp Sensor which helps us collect the data about the temperature and accordingly turn on the ***fan*** when the temperature is beyond a certain degree. Here, we have set the temperature limit to ***40 degree*** Celcius beyond which the fan gets turned on automatically without any manual operation.
3. **Automated Light Bulb Operator:**
   In this system, we use a Light detecting sensor named ***KY-018 LDR*** which detects the amout of darkness in the room. We select the maximum darkness range to ***300*** beyond which the ***LED*** gets turned on automatically.
4. **Smoke detection Alarm:**
   Here, we use a smoke detecting sensor named ***MQ2*** Gas Sensor which detects the presence of smoke in the room hence alarming us about accidental fire in the room. We set the limit upto ***500*** ppm beyond which the ***buzzer*** starts ringing making us aware of any unwanted smoke or fire in the room.

Circuit Diagram


Dist Sensor
Temp Sensor
LDR
Gas Sensor
Breadboard
Arduino
Buzzer
Fan
LED
Buzzer
VCC-5V
Trig-D5
Echo-D2
Gnd
VCC-5V
Gnd
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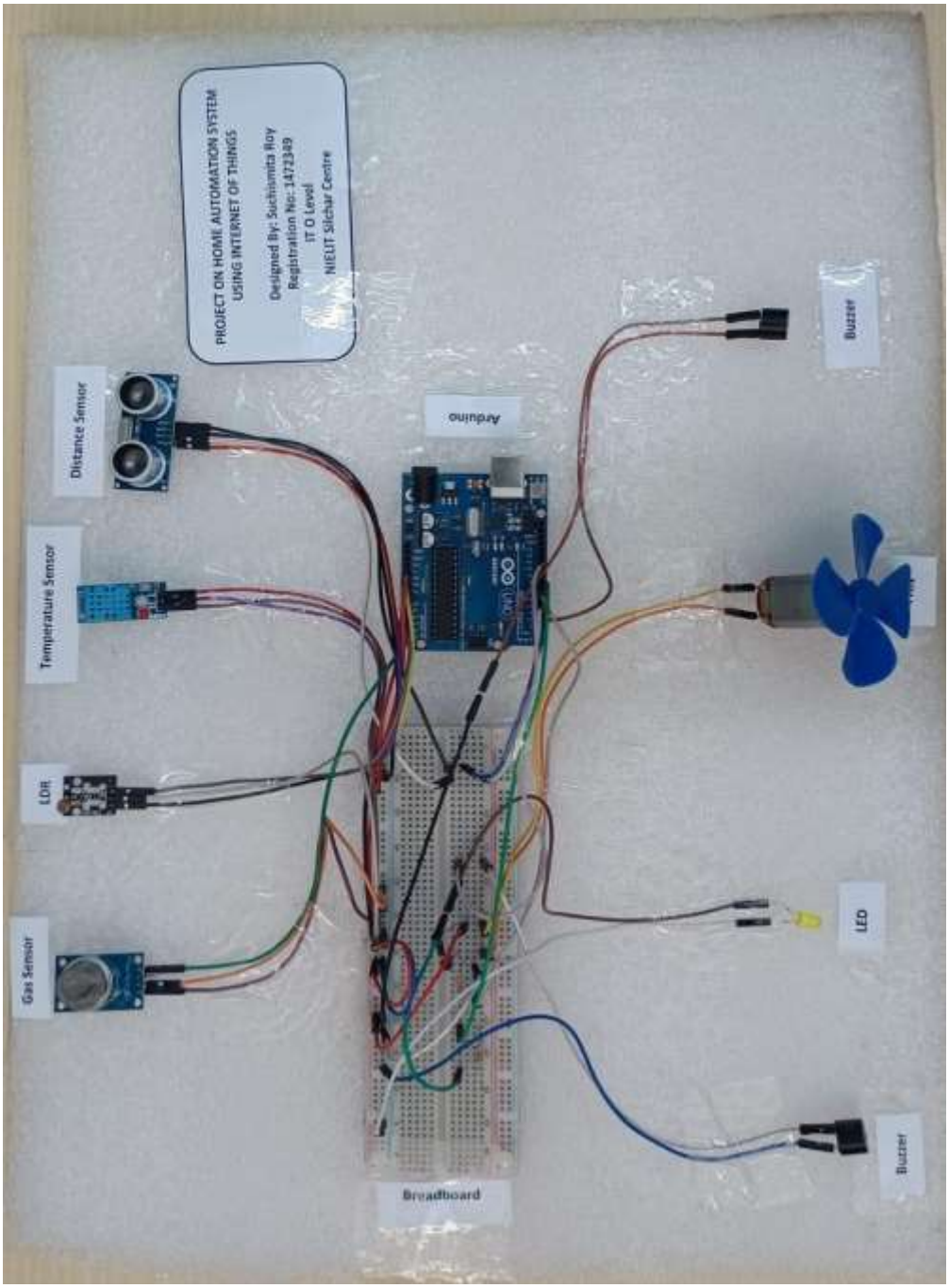
Home Automation System using Internet of Things



Code Used in Arduino IDE

```
#define BUZZER_PIN 11
long duration;
int distance;
int x;
int val;
int fan=7;

 void setup()
{
  pinMode(5,OUTPUT);
  pinMode(2,INPUT);
  pinMode(4,OUTPUT);
  pinMode(A3,INPUT);
  pinMode(8,OUTPUT);
  pinMode(7,OUTPUT);
  pinMode(A0,INPUT);
  pinMode(BUZZER_PIN, OUTPUT);

 Serial.begin(9600);
}
void loop()
```

```
{
  //Distance Sensor
  digitalWrite(5,LOW);
  delayMicroseconds(2);
  digitalWrite(5,HIGH);
  delayMicroseconds(10);
  digitalWrite(5,LOW);

  duration=pulseIn(2,HIGH);
  distance=duration*0.034/2;

  if(distance<20)
   {digitalWrite(4,HIGH);
    delay(1000);
    digitalWrite(4,LOW);
    delay(1000);
    }
  else
   {
    digitalWrite(4,LOW);
    }

   //LDR
  digitalWrite(A3,HIGH);
```

```
 x=analogRead(A3);
 Serial.print("Darkness: ");
Serial.println(x);
  if (x>300)
   {
  digitalWrite(8,1);
   Serial.println("The bulb is on");


   }
  else
   {
    digitalWrite(8,0);
   Serial.println("The bulb is off");
   }


  //Temperature Sensor
  digitalWrite(A0,HIGH);
  delay(1000);
  digitalWrite(A0,LOW);
  val=analogRead(A0);
  float nv=(val/1024.0)*500;
  float cel=nv/10;
```

```
  if(cel>30)
  {
    digitalWrite(fan,HIGH);
  }
  else
  {
    digitalWrite(fan,LOW);
    }

//Smoke Detector
  int sensorValue = analogRead(A5);
  if (sensorValue >500)
  {
    analogWrite(BUZZER_PIN, 50);
  }
  else
  {
    analogWrite(BUZZER_PIN, 0);
  }
  delay(1000);


Serial.print("Temperature: ");
Serial.print(cel);
```

```
Serial.println("C");
Serial.print("Distance: ");
Serial.print(distance);
Serial.println("cm");
Serial.println("Smoke Level: ");
Serial.println(sensorValue);
Serial.println(“     “);
Serial.println(“     “);
delay(1000);

}
```

<u>We can monitor the data of the parameters in the Serial Monitor</u>

Future Scope And Further Enhancements

With the advancements in IoT and AI technology, smart appliances and gadgets are able to mimic daily routines and this is just the beginning. In a few years, the future of home automation will transform the way humans interact with all the appliances and the way they will work together with other appliances.

With the increased inter-connectivity of devices, it's not new to imagine a smart home which will anticipate one's needs and offer consumers a different lifestyle that is more convenient and smarter.

The scope for home automation is increasing in India as it not only provides convenience to the individuals but also encapsulates the emerging digital India. The easily accessible and cost-effective smart home products are being available for everyone to use and will continue to get even cheaper with the rising competition and increasing demands.

Overall, the future of home automation looks to be one of increased convenience, efficiency, and personalization, as technology continues to advance and become more integrated into our homes.

Moreover, the same idea can be used in many other fields such as Commercial Systems, Factories, Educational Institutions, Restaurants and even in Transport Vehicles and many more.

Thank You